EBERHARD KARLS
UNIVERSITÄT
TÜBINGEN

SFB 833
The Construction of Meaning

Andreas Kehl
Linguistic Evidence 2018

Project A7: Focus and Extraction in Complex Constructions and Islands

Adjunct islands and aspectual classes

Experimental evidence

1 Theoretical background

Adjunct clauses are traditionally classified as strong islands under syntactic approaches to locality (Huang's 1982 CED). Most current versions of phrase structure converge on the opacity of adjuncts.

1.1 Exceptions to adjunct islands

- Early examples of apparent exceptions to the Adjunct Condition of Huang's CED
- Notably in Demonte (1988) and Borgonovo & Neeleman (2000)
  → not all adjuncts appear to be absolute islands
- Different approaches to counterexamples:
  - Interaction with grammatical verb types (unergative / unaccusative)
    → Borgonovo & Neeleman (2000)
  - Interaction with aspectual verb classes and agentivity
    → Truswell (2007, 2011)

1.2 Aspectual classes determine transparency (Truswell 2007)

- Truswell (2007) relates the transparency of non-prepositional present participle adjunct clauses to the aspectual classes (telicity) of the predicates involved.
- **Central claim:** only atelic adjunct predicates modifying a telic matrix predicate allow extraction out of the adjunct (Truswell 2007:1369).
- **Predictions:**
  - Declarative sentences with a telic adjunct predicate are less acceptable than sentences with an atelic adjunct predicate.
  - If the adjunct predicate is atelic, the telicity of the matrix predicate does not influence acceptability in declaratives.
  - The effect of *wh*-extraction (island effect) is smallest for the combination [telic/atelic], resulting in (marginal) acceptability.

2 Experiment

This experiment tests the predictions made in Truswell (2007) with a rating study; the four combinations of aspectual predicate classes are compared to each other both in declaratives and under *wh*-extraction. The hypotheses are derived from the discussion in Truswell (2007).

2.1 Study design

Design: 2x2 factorial design with the factors MATRIX AKTIONSART (levels: telic / atelic), ADJUNCT AKTIONSART (levels: telic / atelic), and the between-subjects factor STRUCTURE (levels: base position / *wh*-extraction).

Participants: 85 native speakers of English recruited via Mechanical Turk; 42 saw the items in base position (1), 43 with *wh*-extraction (2).

Sample items:

(1) **Base**
- a. Liz | **got nervous | noticing the large spider on her arm**. [**telic**/**telic**]
- b. Liz | **got nervous** | *watching a horror movie*. [**telic**/*atelic*]
- c. Liz | *lay in bed* | **noticing the large spider on her arm**. [*atelic*/**telic**]
- d. Liz | *lay in bed* | *watching a horror movie*. [*atelic*/*atelic*]

(2) ***wh*-extraction**
- a. (*) What did Liz | **get nervous | noticing** _? [**telic**/**telic**]
- b. What did Liz | **get nervous** | *watching* _? [**telic**/*atelic*]
- c. (*) What did Liz | *lie in bed* | **noticing** _? [*atelic*/**telic**]
- d. (*) What did Liz | *lie in bed* | *watching* _? [*atelic*/*atelic*]

2.2 Hypotheses (based on Truswell 2007)

(3)
- a. AKTIONSART of the adjunct predicate affects the acceptability of declarative sentences → conditions 2 and 4 should receive significantly better ratings than conditions 1 and 3 in base position.
- b. AKTIONSART of the matrix predicate does not affect the acceptability of declarative sentences if the adjunct predicate is atelic (→ conditions 2 and 4 should receive similar ratings in base position).
- c. There is an interaction between MATRIX AKTIONSART and STRUCTURE for conditions 2 and 4.
- d. STRUCTURE has a significant effect on all conditions (island effect → all four conditions should receive lower ratings under *wh*-extraction compared to base position).

2.3 Results

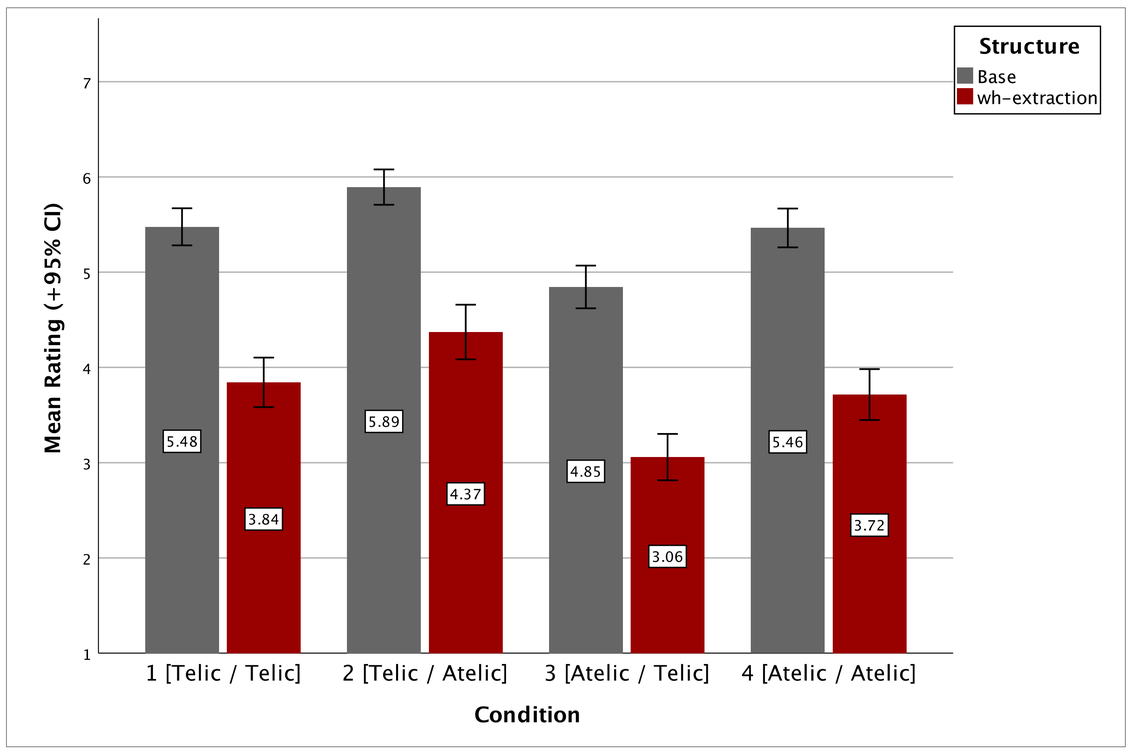


Figure 1: *Results of a rating study with the factors* MATRIX AKTIONSART, ADJUNCT AKTIONSART, *and the between-subjects factor* STRUCTURE.

2.4 Main findings

- Hypothesis (3a) is confirmed (highly significant difference)
- Hypothesis (3b) is **not** confirmed (significant difference between conditions 2 & 4 in base position) [$t(41) = 2.828$, $p < 0.01$]
- Hypothesis (3c) is **not** confirmed (interaction is not significant)
  [$F_1(1) = 1.041$, $p = 0.311$, $\eta^2 = 0.012$; $F_2(1) = 0.483$, $p = 0.497$, $\eta^2 = 0.031$]
- Hypothesis (3d) is confirmed by the effect of factor STRUCTURE
  [$F_1(1) = 122.228$, $p < 0.001$, $\eta^2 = 0.596$; $F_2(1) = 90.858$, $p < 0.001$, $\eta^2 = 0.866$]

3 Discussion

- Confirmation of hypothesis (3a) shows that there are restrictions on Aktionsart combinations in base position → the pattern is not entirely predicted by Truswell (2007).
- Disconfirmation of (3b) shows that the Aktionsart of the matrix predicate is significant when the adjunct predicate is atelic → declaratives are not as unconstrained as Truswell (2007) claims.
- Lack of interaction between conditions 2 & 4 (hypothesis 3c) shows that the main claim of Truswell (2007) does not show up in the experiment
  → *wh*-extraction is not responsible for the difference in acceptability since the two conditions are affected equally by extraction.
- The similar pattern for both levels of STRUCTURE suggests that there are general limitations on the combination of two predicates in the construction; the observed difference in acceptability under extraction is a 'conspiration' of independent, linear-additive violation costs.

4 Planned experiments

- Agentivity (Truswell 2011), grammatical verb type (Borgonovo & Neeleman 2000), context effects, filled-gap effects (transitivity) / processing.
- *I already know that John came home noticing a black cat in the garden last Saturday, but what did he come home noticing* _ TODAY*?*

References

Borgonovo, C., and A. Neeleman. 2000. "Transparent adjuncts." *Canadian Journal of Linguistics / La Revue Canadienne de Linguistique* 45(3/4). 199–224.
Demonte, V. 1988. "Remarks on secondary predicates: C-command, extraction, and reanalysis." *The Linguistic Review* 6. 1–39.
Huang, C.-T. J. 1982. *Logical relations in Chinese and the theory of grammar.* Ph.D. dissertation, MIT.
Truswell, R. 2007. "Extraction from adjuncts and the structure of events" *Lingua* 117. 1355–1377.
Truswell, R. 2011. *Events, phrases, and questions.* Oxford: OUP.

Universität Tübingen · SFB 833 The Construction of Meaning
A7 Focus and Extraction in Complex Constructions and Islands
Nauklerstraße 35 / 3.07 · 72074 Tübingen · Germany
Telefon +49 7071 29-77158 · andreas.kehl@uni-tuebingen.de